FDS EXPERIMENT-7

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Random Sampling and Sampling Distribution.

```
import numpy as np
from itertools import combinations
#Random Sampling Distribution of the mean
```

```
n = 2
population = np.array([40,45,50,55,60])

print("Sample size :",n)
print("Population Data :",population)
```

```
Sample size : 2
Population Data : [40 45 50 55 60]
```

```
population_mean = population.mean()
print("Population Mean: ",population_mean)
```

```
Population Mean:  50.0
```

```
samples = [
    tuple(int(x) for x in sample)
    for sample in combinations(population,n)
]


print("Samples:")
print(list(samples))
```

```
Samples:
[(40, 45), (40, 50), (40, 55), (40, 60), (45, 50), (45, 55), (45, 60), (50, 55), (50, 60), (55, 60)]
```

```
sample_means = np.array([np.mean(x) for x in samples])
print("Samples Mean:")
print(sample_means)
```

```
Samples Mean:
[42.5 45.  47.5 50.  47.5 50.  52.5 52.5 55.  57.5]
```

```
sampling_distribution = np.mean(sample_means)
print("Mean of the sampling distribution :",sampling_distribution)
```

```
Mean of the sampling distribution : 50.0
```

```
population_std = np.std(population)
print("Standard Deviation of population :",population_std)
```

```
Standard Deviation of population : 7.0710678118654755
```

```
standard_error = population_std/(n**0.5)
print("Standard Error :",standard_error)
```

```
Standard Error : 5.0
```